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POLICY AT A GLANCE

Breed Adaptation, Feed Availability, and Disease Management to Achieve Sustainable Milk Production

Agriculture and Allied Sectors in India

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The '**Policy-at-a-Glance**' provides users with curated information for first-level analysis and inter-state comparisons of policy. The document brings together all topic-relevant national and state-level acts, policies, and schemes in one place. Further, it identifies for a reader what policy features are covered in which document in a simple table.

The first section of this document sets out some '*Context and Background*' on the topic; the subsequent sections explain the '*Policy Features*' that have been compared and include a '*Glossary*' of technical terms used; the last section is the 'Table of Insights' that marks out what '*Policy Feature*' has been covered (and on which page) in each of the topic-relevant documents.

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1. Context

Milk and dairy products are valuable components of India's food culture, making significant contributions to food and nutrition security nationally. However, despite being the world's largest producer of milk, the country's yield per milch animal remains low in comparison to the global average. Low livestock productivity in India is a result of inadequate nutrition and limited coverage of planned breeding programmes, as well as larger structural issues such as socioeconomic inequality (Gowane et al., 2019). With the onset of climate change, heat stress – which leads to reduced appetite, lower weight gain and decreased fertility in cattle – is expected to further negatively impact milk productivity. By 2085, the milk output of India is projected to reduce by nearly 25% from that of 2005 (Thornton et al., 2022), and such a significant decline could be detrimental for India's food security. Enhancing the productivity of milch animals through adequate feed and fodder, last-mile implementation of breeding services, and livestock health management are crucial for ensuring sustainable milk production.

This document compiles information regarding various policy approaches aimed at ensuring sustainable milk production in India. As the dairy landscape in India undergoes continuous evolution, the instrumental role of state governments in shaping sector-specific policies and influencing the livelihoods of the workforce becomes apparent. A comprehensive understanding of the diverse approaches adopted by different states is imperative for fostering equitable and sustainable development within the sector.

2. Policy Features

This section identifies a common set of policy features that documents referred to in Section 4 (Table of Insights) cover. The features are classified under the following three broad aspects of policy: 1) breed adaptation; 2) feed management; and 3) extension services.

1. Breed Adaptation

- **Selective breeding and crossbreeding practices:** The low productivity of indigenous animals has been highlighted as a key challenge to sustainable milk production in India, with experts highlighting the need for structured breeding programmes. At the same time, many indigenous animals have been found to be more tolerant to heat stress as compared to exotic milch animals. To this end, standardising, supporting, and facilitating selective breeding and crossbreeding practices to retain indigenous characteristics through policies allows for the development of high-yielding livestock with increased tolerance to current and future climate stressors. Between 2012 and 2019, there was a 34% increase in the number of crossbred cows in India even as the number of indigenous cattle stagnated (Department of Animal Husbandry and Dairying, 2022). Gujarat's [Livestock \(Bovine\) Breeding Policy, 2018](#) identifies specific local breeds that are known to be resilient to climate change and outlines the technical standards for crossbreeding practices. This aspect of promoting, standardising, and regulating selective breeding and crossbreeding practices is included in policies at the national level as well as in the policies of several other states/UTs.
- **Artificial insemination:** The availability of scientific breeding technologies is key to better breed adaptation. Artificial insemination is one such tried-and-tested method. Policies related to artificial insemination include those that establish minimum standards and norms, as well as those that promote and facilitate the use of technology in the local context. Assam's [State Cattle and Buffalo Breeding Policy \(2020\)](#) is one such policy which provides impetus for large-scale utilisation of artificial insemination and engagement of breeders' association to promote the same. National-level policies and the policies of multiple other states/UTs have similar provisions to promote and regulate artificial insemination technologies and processes for livestock.

2. Feed Management

- **Fodder cultivation and management of pastoral systems:** Lack of nutritious fodder, both green and dry, is a major challenge to sustainable milk production. The promotion of fodder cultivation through incentives and initiatives designed for the local context will ensure that livestock have adequate feed, thereby leading to higher productivity of milch animals. These include policies on the management of hortipasture, silvopasture, and common property resources. The centrally-sponsored [Fodder and Feed Development Scheme](#) is one such policy initiative which promotes the cultivation of superior varieties of fodder and advocates the adoption of technology for fodder conservation through seed distribution, establishment of infrastructure (e.g., fodder block making units, feed testing laboratories, etc.), and demonstration of techniques and technologies. Such aspects are covered in the policies of a majority of the states/UTs, mostly through state-level action plans.
- **Technologies for fodder production and conservation:** The integration of technology during the production and post-harvest stages of fodder cultivation is expected to significantly improve the quality and quantity of the feed available to livestock, which in turn will enable sustainable milk production. Action plans and initiatives of numerous states/UTs, in addition to central policies, promote the uptake of technologies for fodder production and conservation. For instance, the Fodder Resources Development Plan for Gujarat calls for demonstrations of technologies for fodder conservation such as fodder blocks, silage, and leaf meal preparation.
- **Quality control of livestock feed:** The quality of livestock feed is a major determinant of the health and productivity of livestock. Evidence from India and other countries around the world suggests that higher quality feed results in higher productivity of milch animals. In particular, the impaired quality of livestock feed in India has been identified as a significant challenge to sustainable milk production. Policies regulate the quality of such feed by defining minimum standards and establishing monitoring systems. Rajasthan's [State Livestock Development Policy](#) demands for an institutional arrangement to monitor the production and distribution of quality cattle feed in addition to conducting tests to ensure the quality of marketed feed and fodder. This aspect of enabling quality control of livestock feed is present in national-level policies as well as in that of several other states/UTs.

3. Extension Services

- **Community-based livestock breeding and health services:** The livestock sector in India predominantly consists of small and marginal farmers in rural areas, most of whom own between one to three animals. In this scenario, it is essential that breeding and health services, including first aid and vaccinations, are accessible and affordable at the community level. This ensures disease-free livestock that provide higher yields of milk. The current reach of livestock vaccination is inadequate as demonstrated by the number of animals vaccinated for lumpy skin disease (LSD) since January 1st, 2022. Of the 27.7 crore animals registered with the Livestock Health and Disease Control Programme, just over 2.5 crore animals have received LSD vaccinations (National Dairy Development Board, n.d). Andhra Pradesh's [Gopalmitra initiative](#) enables community-based breeding services by deploying 'Gopalmitras,' a trained cadre providing artificial insemination, first aid and vaccination services at the village level. Central policies and those of a number of other states/UTs have similar schemes or policies that allow for community-based livestock breeding and health services.

3. Glossary

- **Dairy animals:** According to Food and Agriculture Organization (FAO), the world's milk production is almost entirely derived from cattle, buffaloes, goats, sheep and camels. Other less common milk-producing animals are yaks, horses, reindeers and donkeys (FAO, n.d.). In India, the most common dairy animals are buffaloes (45% of total milk production); cross-bred, non-descript and indigenous cows (52%); and goats (3%) (Department of Animal husbandry and Dairying, 2022).
- **Indigenous/non-descript breeds:** Indigenous animals are those that belong to descript/non-descript breeds of indigenous origins, while non-descript breeds are those indigenous breeds which cannot be classified as a particular recognised breed or do not have more than 50% similarities with any recognised breed (Department of Animal Husbandry and Dairying, 2019).
- **Exotic breeds:** Exotic breeds are animals which are developed in other countries, e.g., Jersey and Holstein Friesian cows (Department of Animal Husbandry and Dairying, 2019).

4. Table of Insights

Geography	Data	Policy	Aspects: Breed Adaptation		Aspects: Feed Management			Aspects: Extension Services
			Selective breeding and sustainable crossbreeding practices	Artificial insemination	Fodder cultivation and management of pastoral systems	Technologies for fodder production and conservation	Quality control of livestock feed	
	Per capita availability of milk in grams per day (National Dairy Development Board)	Reference document						
Centre	444	National Livestock Policy	✓ (p. 13)	✓ (p. 13)	✓ (p. 19)			
		National Livestock Mission	✓ (p. 18)	✓ (p. 18)	✓ (p. 27)		✓ (p. 4)	✓ (p. 21)
		National Programme for Bovine Breeding and Dairy Development		✓ (p. 2)				
		Rashtriya Gokul Mission	✓ (p. 1)	✓ (p. 1)				✓ (p. 8)
		Rashtriya Gokul Mission: Revised Operational Guidelines	✓ (p. 10)	✓ (p. 12)				
		Fodder and Feed Development Scheme			✓ (p. 1)	✓ (p. 1)	✓ (p. 1)	
		Accelerated Fodder Development Programme			✓ (p. 5)	✓ (p. 5)		
		Livestock Health and Disease Control Scheme						✓ (p. 18)
		National Animal Disease Control Programme for Foot and Mouth Disease and Brucellosis						✓ (p. 15)
Andhra Pradesh	799	Andhra Pradesh Bovine Breeding Act, 2021		✓ (p. 13)				
		Andhra Pradesh Animal Feed (Regulation of Manufacture, Quality Control, Sale and Distribution) Act, 2020					✓ (p. 13)	
		Andhra Pradesh Livestock Development Agency: Artificial Insemination	✓	✓				✓
		Fodder Resources Development Plan for Andhra Pradesh			✓ (p. 13)	✓ (p. 13)	✓ (p. 13)	
		Integrated Livestock Management (Scheduled Caste sub Plan)			✓ (p. 13)		✓ (p. 13)	
Arunachal Pradesh	82	Arunachal Pradesh Livestock and Poultry Breeding Policies	✓ (p. 1)					
Assam	77	State Cattle and Buffalo Breeding Policy	✓ (p. 14)	✓ (p. 18)				
		Fodder Resource Development Plan for Assam			✓ (p. 23)	✓ (p. 30)	✓ (p. 30)	
Bihar	270	Bihar Animal Breeding Policy		✓ (p. 2)				
		Fodder Resource Development Plan for Bihar			✓ (p. 26)	✓ (p. 38)	✓ (p. 33)	
Chandigarh	129	Chandigarh Breeding Policy	✓	✓				

Geography	Data	Policy	Aspects: Breed Adaptation		Aspects: Feed Management			Aspects: Extension Services
	Per capita availability of milk in grams per day (National Dairy Development Board)	Reference document	Selective breeding and sustainable crossbreeding practices	Artificial insemination	Fodder cultivation and management of pastoral systems	Technologies for fodder production and conservation	Quality control of livestock feed	Community-based livestock breeding and health services
Chhattisgarh	172	Chhattisgarh Livestock Development and Breeding Policy	✓ (p. 17)	✓ (p. 9)				
Goa	111	Goa Breeding Policy for Cattle and Buffaloes	✓ (p. 1)					
		Scheme for Incentive to Green Fodder Cultivation for Perennial and Seasonal Fodder Development			✓ (p. 1)			
		Breeding and Rearing of Cows/Buffaloes / Calves Scheme		✓ (p. 2)				
		Fodder Resources Development Plan for Goa			✓ (p. 23)	✓ (p. 31)	✓ (p. 30)	
Gujarat	656	Livestock (Bovine) Breeding Policy	✓ (p. 2)	✓ (p. 2)				
		Scheme: Incentive to Owners of A.I. Born Female Calf through Pure Breeding in Indigenous Cow	✓					
		Model Fodder Development and Seed Production Programme			✓			
Haryana	1,081	Bovine Breeding Policy	✓ (p. 1)					
		Scheme for Conservation and Development of Indigenous Cattle and Murrah Development		✓ (p. 1)				
Himachal Pradesh	599	Breeding Policy for Cattle and Buffaloes	✓ (p. 5)	✓ (p. 4)				
Jammu and Kashmir	557	The Jammu and Kashmir Breeding (Regulation of Production, Sale of Bovine Semen and Artificial Insemination Services) Act, 2018		✓ (p. 10)				
		Jammu and Kashmir State Livestock Breeding Policy	✓ (p. 5)	✓ (p. 5)				
		Fodder Resources Development Plan for Jammu and Kashmir			✓ (p. 27)	✓ (p. 27)	✓ (p. 48)	
		Feed and Fodder Development Scheme			✓			
Jharkhand	187	Jharkhand Livestock Breeding Policy	✓ (p. 3)	✓ (p. 3)				
		Fodder Resources Development Plan for Jharkhand			✓ (p. 30)	✓ (p. 35)		
Karnataka	483	Karnataka Breeding Policy		✓ (p. 2)				
		Fodder Development Scheme			✓ (p. 5)			

Geography	Data	Policy	Aspects: Breed Adaptation		Aspects: Feed Management			Aspects: Extension Services
	Per capita availability of milk in grams per day (National Dairy Development Board)	Reference document	Selective breeding and sustainable crossbreeding practices	Artificial insemination	Fodder cultivation and management of pastoral systems	Technologies for fodder production and conservation	Quality control of livestock feed	Community-based livestock breeding and health services
Kerala	196	Kerala State Breeding Policy		✓ (p. 4)				
		Fodder Resource Development Plan for Kerala			✓ (p.25)	✓ (p. 34)	✓ (p. 33)	
		Scheme: Production and Conservation of Fodder			✓			
Madhya Pradesh	616	State Livestock Breeding Policy	✓ (p. 1)	✓ (p. 1)				
		Improved Cattle Breeding Scheme	✓					
Maharashtra	315	Maharashtra State Livestock Policy	✓ (p. 1)	✓ (p. 5)				
		Policy for Feed, Fodder, Animal Nutrition and Grazing			✓	✓		
		Livestock Policy for Health Care						✓
		Fodder Resources Development Plan for Maharashtra			✓ (p. 29)	✓ (p. 44)	✓ (p. 42)	
Manipur	65	North Eastern Council Guidelines	✓ (p. 39)	✓ (p. 97)	✓ (p. 93)			
		Manipur State Breeding Policy	✓ (p. 3)	✓ (p. 2)				
		Fodder Resources Development Plan for Manipur			✓ (p. 24)	✓ (p. 33)	✓ (p. 30)	
Meghalaya	75	Meghalaya Milk Mission			✓ (p. 12)			
		Meghalaya Bovine and Pig Breeding Policy	✓ (p. 70)	✓ (p. 74)				
		Fodder Resources Development Plan for Meghalaya			✓ (p. 25)	✓ (p. 34)		
Mizoram	55	Fodder Resources Development Plan for Mizoram			✓ (p. 25)	✓ (p. 34)	✓ (p. 33)	
		North Eastern Council Guidelines	✓ (p. 39)	✓ (p. 97)	✓ (p. 93)			✓ (p. 91)
Nagaland	76	Fodder Resources Development Plan for Nagaland			✓ (p. 25)	✓ (p. 33)	✓ (p. 32)	
		North Eastern Council Guidelines	✓ (p. 39)	✓ (p. 97)	✓ (p. 93)			✓ (p. 91)

Geography	Data	Policy	Aspects: Breed Adaptation		Aspects: Feed Management			Aspects: Extension Services
	Per capita availability of milk in grams per day (National Dairy Development Board)	Reference document	Selective breeding and sustainable crossbreeding practices	Artificial insemination	Fodder cultivation and management of pastoral systems	Technologies for fodder production and conservation	Quality control of livestock feed	Community-based livestock breeding and health services
Odisha	144	Scheme: Support to Farmers for Seasonal Fodder Production			✓ (p. 1)			
		Odisha Bovine Breeding Policy	✓ (p. 1)	✓ (p. 2)				
		Scheme: Use of Sex Sorted Semen for Enhancing Milk Production	✓ (p. 1)	✓ (p. 1)				
		Odisha State Livestock Sector Policy	✓ (p. 9)	✓ (p. 10)				✓ (p. 14)
		Odisha Cattle and Poultry Feed (Regulation) Act, 1979			✓ (p. 15)		✓ (p. 4)	
Punjab	1,271	Punjab Bovine Breeding Act, 2016		✓ (p. 63)				
		Fodder Resources Development Plan for Punjab			✓ (p. 12)	✓ (p. 32)	✓ (p. 31)	
		Punjab Animals Feed Stuff and Compound Feed Act, 2016					✓ (p. 3)	
Rajasthan	1,150	Rajasthan State Livestock Development Policy	✓ (p. 15)		✓ (p. 18)		✓ (p. 18)	
		Cattle & Buffalo Breeding Policy	✓ (p. 4)	✓ (p. 5)				
		Fodder Resources Development Plan for Rajasthan			✓ (p. 27)	✓ (p. 32)	✓ (p. 40)	
Sikkim	350	Sikkim Agricultural, Horticultural Input and Livestock Feed Regulatory Act, 2014					✓ (p. 4)	
		Cattle Breeding Policy	✓ (p. 1)	✓ (p. 1)				
		Fodder Resources Development Plan for Sikkim			✓ (p. 27)	✓ (p. 37)	✓ (p. 36)	
Tamil Nadu	362	Tamil Nadu Bovine Breeding Act, 2019		✓ (p. 6)				
		Animal Husbandry Policy Note	✓ (p. 50)		✓ (p. 59)			✓ (p. 70)
Telangana	422	Telangana Live-Stock Improvement Act, 1940						✓ (p. 3)
		Fodder Resources Development Plan for Telangana			✓ (p. 39)	✓ (p. 59)	✓ (p. 55)	
Tripura	146	North Eastern Council Schemes	✓ (p. 1)					✓ (p. 1)
		State Agriculture Infrastructure Development Plan	✓ (p. 36)	✓ (p. 35)	✓ (p. 13)	✓ (p. 30)	✓ (p. 29)	✓ (p. 35)
		State Breeding Policy for Livestock and Poultry	✓ (p. 10)	✓ (p. 12)	✓ (p. 25)	✓ (p. 25)		✓ (p. 36)
		Fodder Resources Development Plan for Tripura			✓ (p. 23)	✓ (p. 31)	✓ (p. 30)	

Geography	Data	Policy	Aspects: Breed Adaptation		Aspects: Feed Management			Aspects: Extension Services
			Selective breeding and sustainable crossbreeding practices	Artificial insemination	Fodder cultivation and management of pastoral systems	Technologies for fodder production and conservation	Quality control of livestock feed	Community-based livestock breeding and health services
Uttar Pradesh	392	Uttar Pradesh Go-Seva Ayog Adhiniyam, 1999			✓ (p. 12)			
		Uttar Pradesh Pashudhan Sudhar Adhiniyam, 1964	✓ (p. 8)					
		Fodder Resources Development Plan for Uttar Pradesh			✓ (p. 24)	✓ (p. 35)	✓ (p. 34)	
Uttarakhand	446	Uttarakhand Bovine Breeding Act, 2018		✓ (p. 17-19)				
		Fodder Resources Development Plan for Uttarakhand			✓ (p. 29)	✓ (p. 46)	✓ (p. 42)	
		Uttarakhand State Livestock Mission	✓ (p. 6)					
		Mukhyamantri Chara Vikas Niti			✓ (p. 9)			
West Bengal	179	West Bengal Breeding Policy	✓ (p. 2)	✓ (p. 3)				
		State Agricultural Plan for West Bengal		✓ (p. 17)	✓ (p. 17, 114)			

Note 1: Documents linked in the tables include Acts, Policies, Schemes, Programmes, and Guidelines which facilitate an overview and comparison of central- and state-level initiatives. While all relevant documents have been identified, the coverage may not be exhaustive. Further, the ‘Policy-at-a-Glance’ is a mapping of these identified documents to highlight aspects of breed adaptation, feed availability, and disease management.

Note 2: Page numbers specified in the Table of Insights are as per the PDF pagination opened in a Google Chrome browser. It is possible to encounter differences in the actual page number when documents are opened in a different web browser/application.

In addition to these national and state-level policies, policies for disease management in livestock have been designed, which play an important role in sustainable milk production. At the national level, the [National Animal Disease Control Programme](#) aims to control FMD by 2025 and eradicate it by 2030. It also aims to lower the incidence of Brucellosis cases in livestock, thereby managing the disease in both animals and humans. Specific strategies include vaccinations, robust surveillance and testing, awareness campaigns, and tracking of individual animals. Similarly, the centrally-driven [Livestock Health and Disease Control Scheme](#) prioritises disease surveillance and vaccination of livestock and poultry, in addition to strengthening veterinary health systems. Priority diseases under this scheme include FMD, Brucellosis, and sheep and goat Plague (Department of Animal Husbandry and Dairying, n.d). The National Animal Disease Reporting System (2.0) is a web-based system that monitors livestock disease in the country. It is complemented by the National Animal Disease Referral Expert System, a remote sensing-enabled geographic information system that collates information on animal diseases and risk factors. It provides monthly livestock disease forecasts for each district and enables immediate action at the state and national levels (Chethan Kumar et al., 2021). At the state level, several states have regulatory Acts in place to prevent and control the spread of contagious diseases in animals, akin to the national [Prevention and Control of Infectious and Contagious Diseases in Animals Act, 2009](#). These legal and policy documents are broad in their coverage with provisions for not just livestock, but also dogs, cats, horses, mules, camels, asses, birds, and bees.

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